Example-1

Quadratic Kernel

$\lfloor u \quad v \rfloor$

u_1, v_1

u_2, v_2

Map Input data from 2-D to 3-D

$\lfloor u \quad v \rfloor$

Vectors Mapping $u, v \rightarrow u, v$

Higher Computational Complexity!!!

$$\left\lfloor \sqrt{2}a_1a_2 \quad \sqrt{2}b_1b_2 \right\rfloor$$
$$a_2^2 \quad b_2^2$$

Normal Model — Calculate $u \quad v$

$$\left\lfloor a_1 \quad a_2 \quad b_1 \right\rfloor$$
$$b_2$$

Kernelized Model — Calculate $u \quad v$

$$\lfloor a_1, a_2, \vee z a_1 a_2, \vee z a_1, \vee z a_2, v_1, v_2 \rfloor$$

Higher Computational Complexity!!!